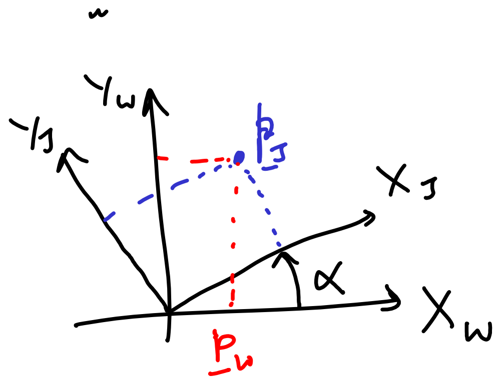
Rotation matrix



$$\underline{p}_W = {}^W R_J \; \underline{p}_J$$

Jetbot w.r.t. world is α

$$\begin{bmatrix} x_W \\ y_W \end{bmatrix} = \underbrace{\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}}_{} \begin{bmatrix} x_J \\ y_J \end{bmatrix}$$

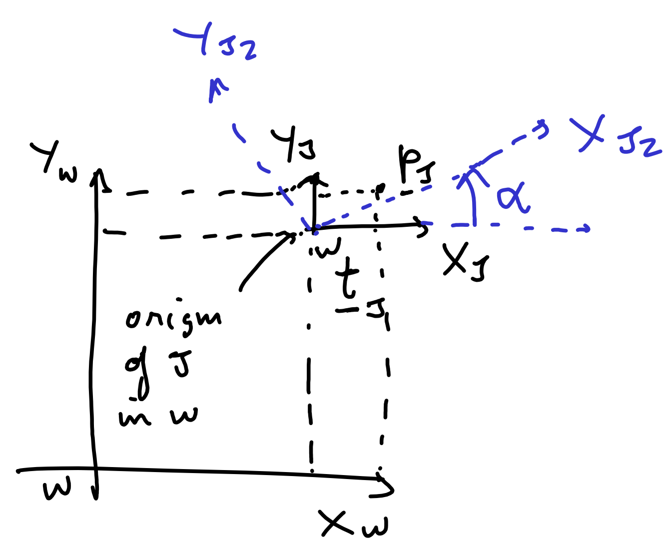
① Orthonormal matrix $\quad {}^W R_J^T \, {}^W R_J = I$

$${}^W R_J(\alpha) \qquad\qquad {}^W R_J^{-1} = {}^W R_J^T = {}^J R_W$$

$${}^W R_J^{-1} = {}^W R_J(-\alpha)$$

② $\det({}^W R_J) = 1$

Rotation + translation



$$\underline{p}_W = \underline{p}_J + {}^W \underline{t}_J$$

$$\underline{p}_W \xleftarrow{{}^W \underline{t}_J} \underline{p}_J \xleftarrow{R(\alpha)} \underline{p}_{J2}$$

① $\underline{p}_J = {}^J R_{J2}(\alpha) \, \underline{p}_{J2}$

② $\underline{p}_W = \underline{p}_J + {}^W \underline{t}_J$

$$\boxed{\underline{p}_W = {}^J R_{J2}(\alpha) \, \underline{p}_{J2} + {}^W \underline{t}_{J2}}$$

Affine transformation

Transformation matrix

$$\underline{p}_w = {}^{J}R_{J_2}\,\underline{p}_{J_2} + {}^{w}\underline{t}_{J_2}$$

$$\begin{bmatrix} x_w \\ y_w \end{bmatrix} = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} x_{J_2} \\ y_{J_2} \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_w \\ y_w \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{J_2} \\ y_{J_2} \\ 1 \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_w \\ y_w \\ \underline{1} \end{bmatrix} = \underbrace{\begin{bmatrix} \cos\alpha & -\sin\alpha & t_x \\ \sin\alpha & \cos\alpha & t_y \\ 0 & 0 & 1 \end{bmatrix}}_{{}^{w}T_{J_2}} \begin{bmatrix} x_{J_2} \\ y_{J_2} \\ 1 \end{bmatrix}$$

homogeneous coordinates

$$\Rightarrow \boxed{\tilde{p}_w = {}^{w}T_{J_2}\,\tilde{p}_{J_2}} \Leftrightarrow \begin{bmatrix} \underline{p}_w \\ 1 \end{bmatrix} = \begin{bmatrix} {}^{w}R_{J_2} & {}^{w}\underline{t}_{J_2} \\ 0^T & 1 \end{bmatrix} \begin{bmatrix} \underline{p}_{J_2} \\ 1 \end{bmatrix}$$

Block-wise matrix multiplications works as long as the component multiplications are <u>valid</u>

Properties of Transformation matrix

① Find the inverse of a transformation matrix using the fact that $R^{-1} = R^T$

$$T = \begin{bmatrix} R_{2\times2} & t_{2\times1} \\ \underline{0}^T_{1\times2} & 1 \end{bmatrix}_{3\times3}$$

$$T^{-1} = ?$$

$$\underline{y} = R\underline{x} + \underline{t}$$

$$\underline{x} = R^T(\underline{y} - \underline{t})$$

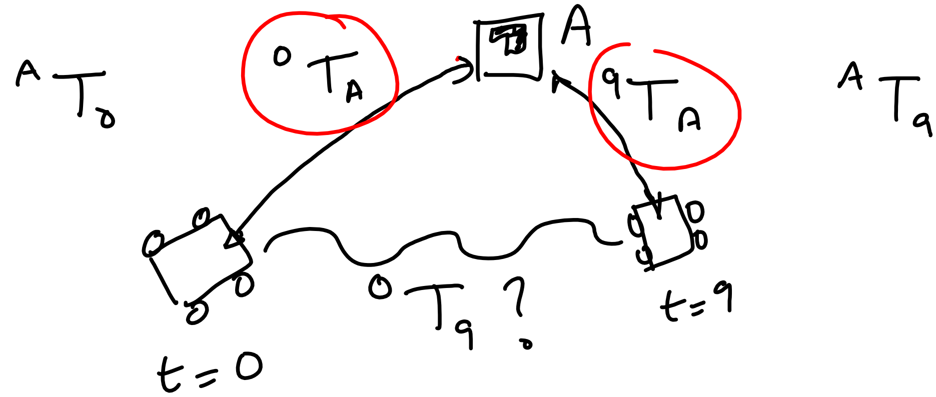
$$\tilde{y} = T\tilde{x}$$

$$\tilde{x} = T^{-1}\tilde{y}$$

$$\underline{x} = R^T\underline{y} - [R^T\underline{t}]_{2\times1}$$

$$\begin{bmatrix} \underline{x} \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} R^T_{2\times2} & -R^T\underline{t}_{2\times1} \\ 0^T_{1\times2} & 1 \end{bmatrix}}_{T^{-1}} \begin{bmatrix} \underline{y} \\ 1 \end{bmatrix}$$

$$\boxed{T = \begin{bmatrix} R & \underline{t} \\ \underline{0}^T & 1 \end{bmatrix} \Rightarrow T^{-1} = \begin{bmatrix} R^T & -R^T\underline{t} \\ \underline{0}^T & 1 \end{bmatrix}}$$



$^0T_9 = ?$

$$\underline{x}_0 = {}^0T_9\, \underline{x}_9$$

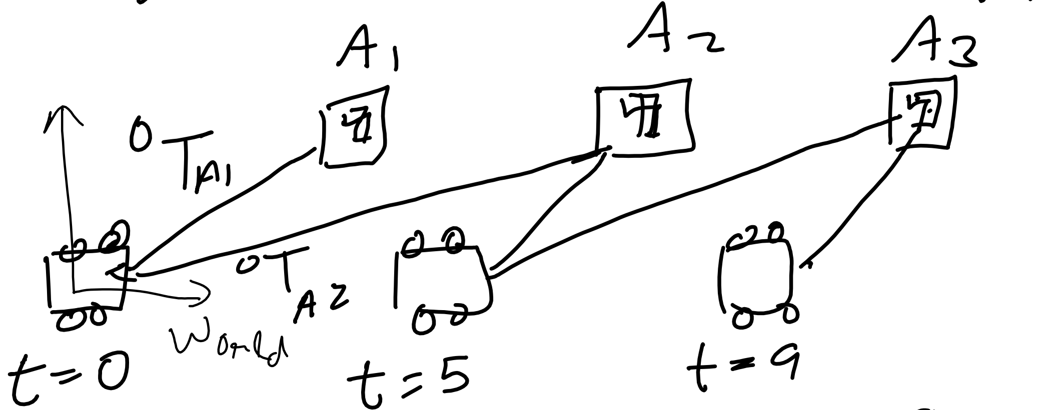
$$\underline{x}_A = {}^AT_9\, \underline{x}_9$$

$$\underline{x}_0 = {}^0T_A\, \underline{x}_A$$

$$\underline{x}_0 = \underbrace{{}^0T_A\, {}^AT_9}_{{}^0T_9}\, \underline{x}_9$$

$\Rightarrow$ $\boxed{{}^0T_9 = {}^0T_A\, {}^AT_9}$ Composition of transformations

Mapping (Landmark based mapping)

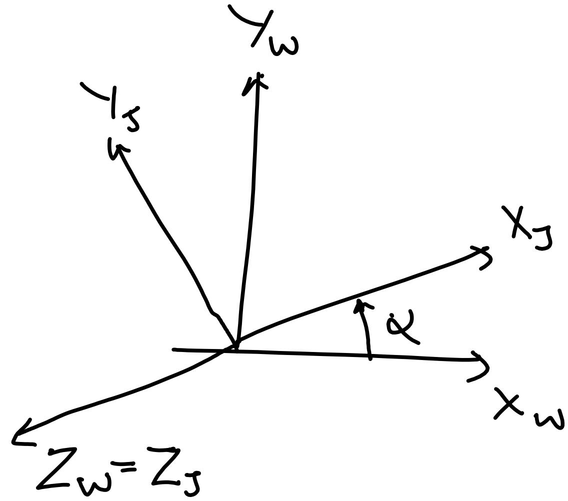


$^0T_{A3} = ?$

$^0T_9 = ?$

$^5T_{A2}$, $^5T_{A3}$, $^9T_{A3}$

3D Rotations



$$\begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} = \underbrace{\begin{bmatrix} \boxed{\begin{matrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{matrix}}^{\text{2D Rot}} & \begin{matrix} 0 \\ 0 \end{matrix} \\ \begin{matrix} 0 & 0 \end{matrix} & 1 \end{bmatrix}}_{R} \begin{bmatrix} x_J \\ y_J \\ z_J \end{bmatrix}$$

① Orthonormal $R^T R = I$

$R^{-1} = R^T$

columns are unit vectors that are $\perp$ to each other

② $\det(R) = 1$

We can rotate in 2D and compose 2D rotations to form 3D rotations.

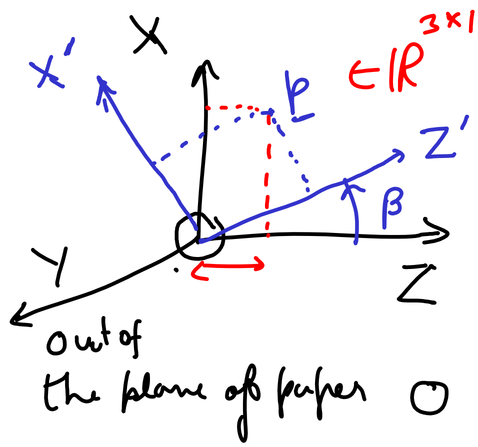
$$\rightarrow \underset{\uparrow \text{around } z\text{-axis}}{R_z(\alpha)} = \begin{bmatrix} R_{2D}(\alpha) & \begin{matrix} 0 \\ 0 \end{matrix} \\ \begin{matrix} 0 & 0 \end{matrix} & 1 \end{bmatrix}_{3\times 3} = R_{X-Y\,\text{plane}}(\alpha)$$

$$\overset{-1}{X\,Y} \rightarrow Z$$

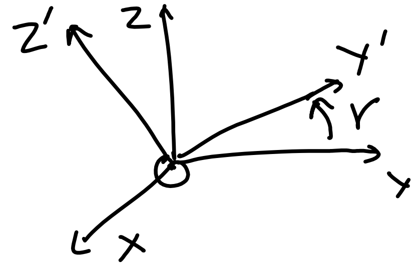
$$\underset{-1}{Z\,X} \rightarrow Y$$

$$R_y(\beta) = \begin{bmatrix} \cos\beta & 0 & +\sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{bmatrix}$$

$$\underset{3\times 1}{p'} = \underbrace{R_y(\beta)}_{3\times 3} \underset{3\times 1}{p}$$



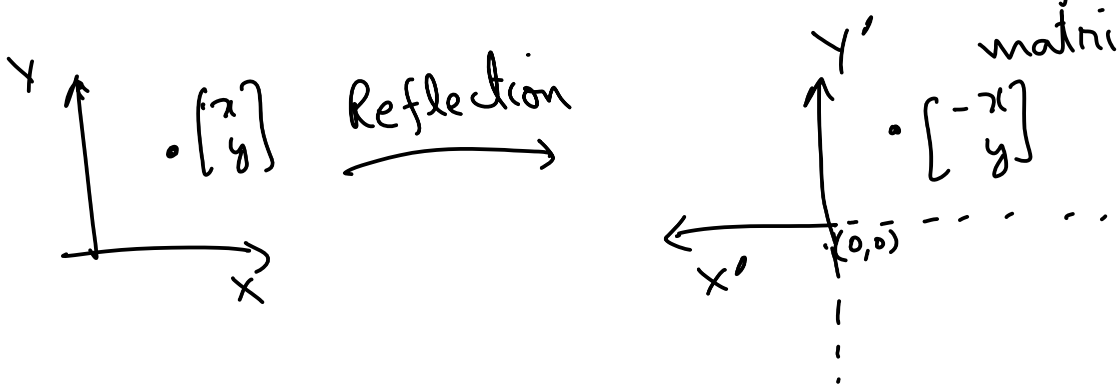
$$R_x(\gamma) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\gamma & -\sin\gamma \\ 0 & \sin\gamma & \cos\gamma \end{bmatrix}$$



. In n-D (2-D, 3-D) any matrix that is Orthonormal and has $\underbrace{\text{determinant} = 1}$ is a valid Rotation matrix

$$U^T U = I \Rightarrow \det(U) = +1 \text{ or } \underbrace{-1}$$

Reflection matrices



$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \underbrace{\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}} \begin{bmatrix} x \\ y \end{bmatrix}$$

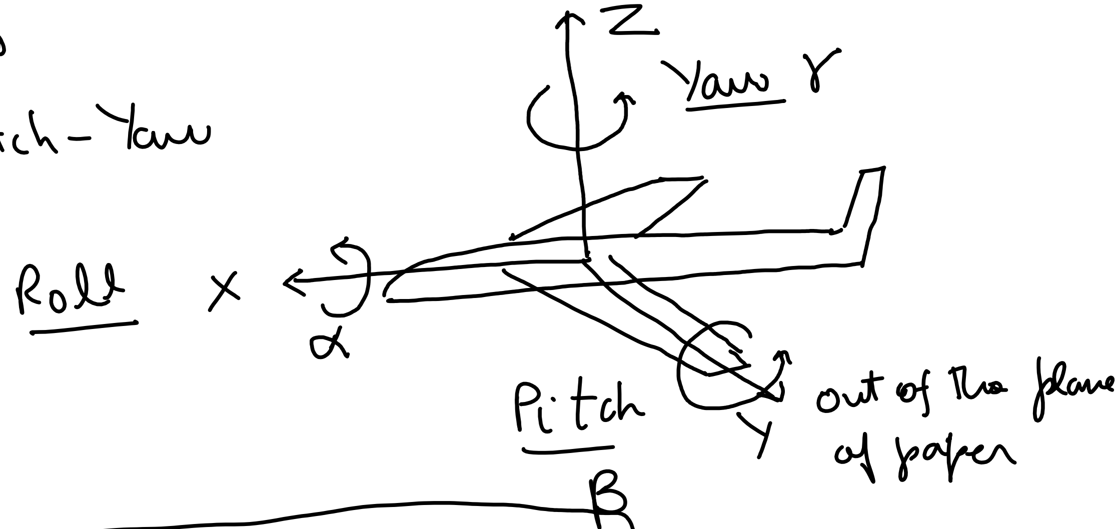
Reflection ≠ Rotation

orthonormal

det( ) = -1

Aeronautics

Roll-pitch-Yaw



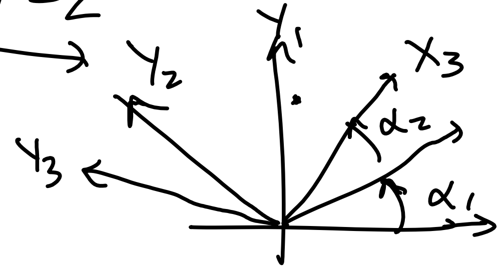
$$\boxed{R_{3D}(\underbrace{\alpha,\beta,\gamma}) = R_x(\alpha)\, R_y(\beta)\, R_z(\gamma)}$$

Euler angles

Axis aligned Rotations

(12 possible Euler angles)

X-Y-Z



$$\underline{p}_2 = {}^2R_3\, \underline{p}_3$$

$$\underline{p}_1 = {}^1\underline{R}_2\, p_2$$

$$\underline{p}_1 = {}^1R_2(\alpha_1)\,{}^2R_3(\alpha_2)$$

roll-pitch-yaw

12 ways

Euler angles : X-Y-Z, Y-Z-X . . . .

$3! = 6$

: X-Y-X, X-Z-X, Y-X-Y, Y-Z-Y

. . . .

6 way

Degrees of Freedom in 3D Rotation matrix

① Orthonormality

② $\det(R) = +1$

DOF

= count the scalars needed to represent a thing

− number of constraints

How many scalars in 3D Rotation matrix = ? = 9 scalars

② $R^T R = I$ ← how many constraints ? $A_{3\times3}\, \underline{x}_{3\times1} = \underline{b}_{3\times1}$

$$\begin{bmatrix} \underline{r}_1^T \\ \underline{r}_2^T \\ \underline{r}_3^T \end{bmatrix} \begin{bmatrix} \underline{r}_1 & \underline{r}_2 & \underline{r}_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

6 constraints

$$\begin{bmatrix} \underline{r}_1^T \underline{r}_1 & \underline{r}_1^T \underline{r}_2 & \underline{r}_1^T \underline{r}_3 \\ \underline{r}_2^T \underline{r}_1 & \underline{r}_2^T \underline{r}_2 & \underline{r}_2^T \underline{r}_3 \\ \underline{r}_3^T \underline{r}_1 & \underline{r}_3^T \underline{r}_2 & \underline{r}_3^T \underline{r}_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

all cols are ⊥

all cols are unit vectors

$$DOF = \underset{\substack{\uparrow \\ \text{free variables}}}{9 \text{ scalars}} - 6 \text{ constraints} = 3 \text{ DOF}$$

② $\det(R) = +1$

$\det(\underset{\substack{\uparrow \\ \text{orthonormal}}}{U}) \in \{+1, -1\}$

Euler angle $X\text{-}Y\text{-}Z$ $\longrightarrow$ Rotation matrix?

α, β, γ $\longleftarrow$ R

$$R = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\alpha & -s_\alpha \\ 0 & s_\alpha & c_\alpha \end{bmatrix}}_{R_x(\alpha)} \underbrace{\begin{bmatrix} c_\beta & 0 & s_\beta \\ 0 & 1 & 0 \\ -s_\beta & 0 & c_\beta \end{bmatrix}}_{R_y(\beta)} \underbrace{\begin{bmatrix} c_\gamma & -s_\gamma & 0 \\ s_\gamma & c_\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{R_z(\gamma)}$$

$c_\alpha = \cos\alpha$

$s_\alpha = \sin\alpha$

$$= \begin{bmatrix} & & \\ & R_x(\alpha) & \\ & & \end{bmatrix} \begin{bmatrix} c_\beta c_\gamma & -s_\gamma c_\beta & s_\beta \\ -s_\gamma & c_\gamma & 0 \\ -s_\beta s_\gamma & +s_\beta s_\gamma & c_\beta \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}}_{R} = \begin{bmatrix} C_\beta C_\gamma & -S_\gamma C_\beta & S_\beta \\ C_\alpha C_\gamma - S_\alpha S_\beta C_\gamma & C_\alpha C_\gamma - S_\alpha S_\beta S_\gamma & -S_\alpha C_\beta \\ S_\alpha S_\gamma - C_\alpha S_\beta C_\gamma & S_\alpha C_\gamma + C_\alpha S_\beta S_\gamma & C_\alpha C_\beta \end{bmatrix}$$

$$\frac{-r_{12}}{r_{11}} = \frac{S_\gamma C_\beta}{C_\gamma C_\beta} = \tan\gamma$$

$$\Rightarrow \gamma = \arctan 2(-r_{12}, r_{11})$$

assuming $C_\beta > 0$

$$\Rightarrow \alpha = \arctan 2(-r_{23}, r_{33})$$

assuming $C_\beta > 0$

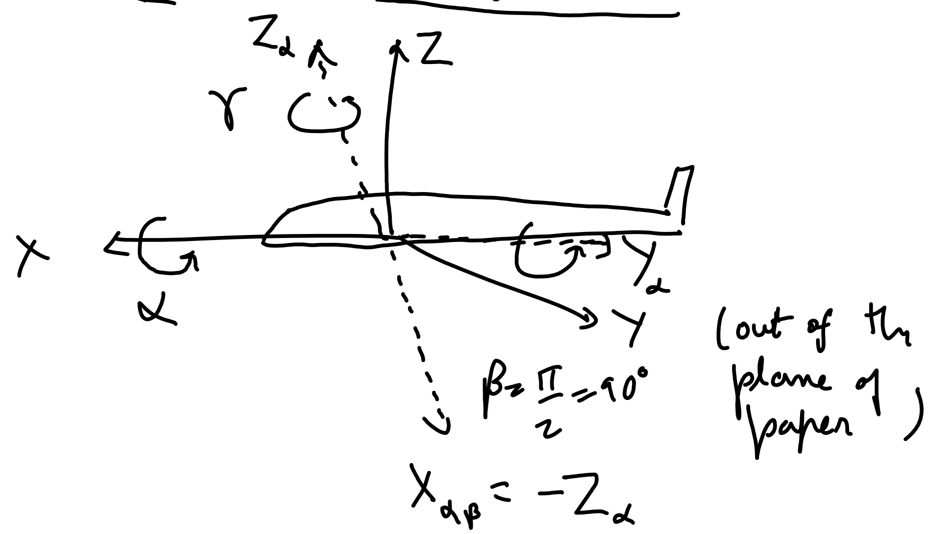
$$C_\beta = \sqrt{r_{23}^2 + r_{33}^2}$$

$$\beta = \arctan 2(r_{13}, C_\beta)$$

$$= \sin^{-1}(r_{13})$$

Rot matrix $\longrightarrow$ Euler angles

Problem with Euler angles : Gimbal lock



α, β, γ $\xrightarrow{\text{unique conversion}}$ Rotation matrix

Euler angles

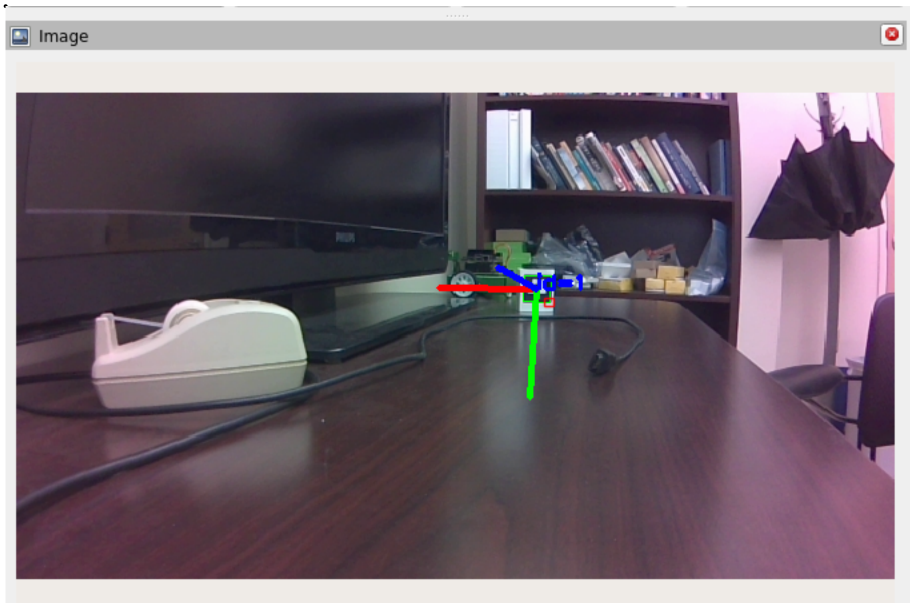
infinite solutions

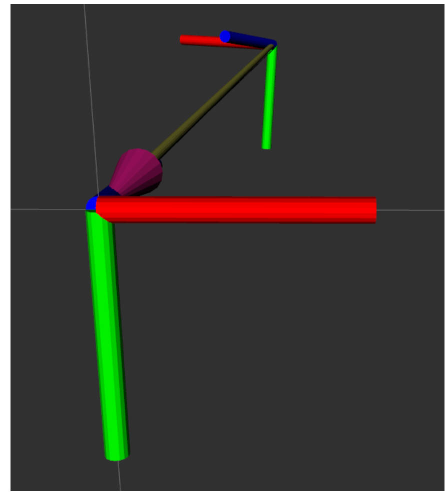
For example

$\alpha = 30°$, $\beta = 90°$, $\gamma = 61°$ $\longrightarrow$ Rotation matrix

$\alpha = 60°$, $\beta = 90°$, $\gamma = 31$

all $\alpha + \gamma = \boxed{91°}$
will lead to the same rotation matrix when $\boxed{\beta = 90°}$





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Publisher count: 1
Subscription count: 0

root@nano-4gb-jp45:/home/jetbot/ece417# ros2 topic echo --once /
aruco_detections
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markers:
- marker_id: 1
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      y: -0.10849999597426438
      z: 0.980432215194246
    orientation:
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      y: 0.9811997541144667
      z: -0.03227342049856023
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boards: []
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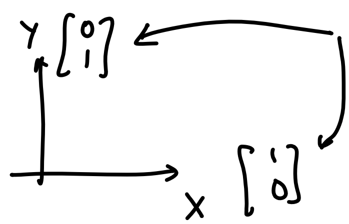
2025-10-22

Basis vectors

3D Rotation Representations
- → Rotation matrics
- → Euler Angles (Problem : Gimbal Lock)
- → Axis - angle
- → Quaternion

Basis vectors

e.g. Coordinate axis



Standard Basis vectors

→ For a given $\overbrace{\text{vector space}}$ basis vectors are a set of vectors that "span" the vector space

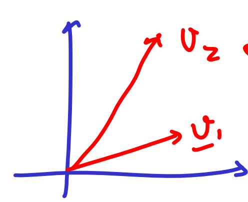
vector space $= \mathbb{R}^2 =$ space of all 2D Real vectors

$\underline{x} = \begin{bmatrix}1\\0\end{bmatrix}$ and $\underline{y} = \begin{bmatrix}0\\1\end{bmatrix}$, $\{\underline{x}, \underline{y}\}$ "spans" $\mathbb{R}^2$

"spans" is the set of all vectors you can obtain by a linear combination of a set of vectors

$\{\underline{x}, \underline{y}\}$ spans $\mathbb{R}^2$ $\quad \mathbb{R}^2 = \{ \underbrace{\alpha \underline{x} + \beta \underline{y}}_{\text{Linear combination}} : \alpha, \beta \in \mathbb{R} \}$

Orthonormal Basis Vectors

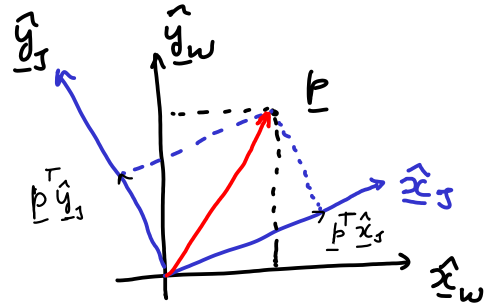


Any two vectors can form a set of Basis vectors as long as they are not parallel.

1. Basis vectors that are mutually $\perp$ (orthogonal / perpendicular)
2. and are unit vectors

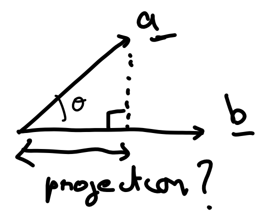
Relationship b/w Orthonormal Basis vectors and Rotation matrices

$\hat{\underline{x}}$ ← hat is used to denote unit vectors



Projection of a vector ?

$$\text{proj}(\underline{a}, \underline{b}) = \frac{\underline{a}^T \underline{b} \leftarrow \text{dot product}}{\|\underline{b}\| \leftarrow \text{magnitude}} = \|a\| \cos\theta$$



$$\underline{p} = (\underline{p}^T \hat{\underline{x}}_J)\, \hat{\underline{x}}_J + (\underline{p}^T \hat{\underline{y}}_J)\, \hat{\underline{y}}_J$$

$$\Rightarrow \underset{2\times1}{\underline{p}} = \underbrace{\begin{bmatrix} \hat{\underline{x}}_J & \hat{\underline{y}}_J \end{bmatrix}_{2\times2}}_{\substack{\text{Basis matrix} \\ B_J}} \underbrace{\begin{bmatrix} \underline{p}^T \hat{\underline{x}}_J \\ \underline{p}^T \hat{\underline{y}}_J \end{bmatrix}_{2\times1}}_{\substack{\text{coordinates} \\ \underline{p}_J}}$$

$$\Rightarrow \underset{2\times1}{\underline{p}} = B_J \underline{p}_J$$

for any set of Basis vectors

$$\underline{p} = B_W \underline{p}_W \quad \text{where} \quad B_W = \begin{bmatrix} \hat{\underline{x}}_W & \hat{\underline{y}}_W \end{bmatrix} \qquad \underline{p}_W = \text{coordinates of } \underline{p} \text{ in } B_W$$

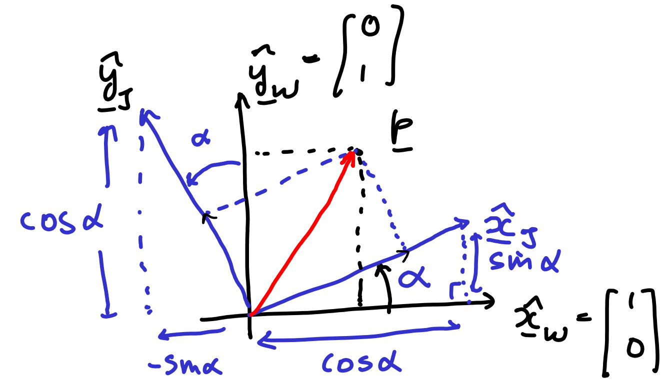
$$\Rightarrow \underline{p} = B_J \underline{p}_J = B_W \underline{p}_W$$

To find $\underline{p}_W$ in terms of $\underline{p}_J$ use orthonormality of B_W

$$\boxed{\underline{p}_W = \underbrace{B_W^T B_J}_{{}^W R_J} \underline{p}_J}$$

Memorize

$$B_w = \begin{bmatrix} \hat{\underline{x}}_w & \hat{\underline{y}}_w \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \underline{I}_{2\times 2}$$



$${}^{w}R_J = ? = B_w^T B_J$$

$$\hat{\underline{x}}_J = \begin{bmatrix} \cos\alpha \\ \sin\alpha \end{bmatrix}, \quad \hat{\underline{y}}_J = \begin{bmatrix} -\sin\alpha \\ \cos\alpha \end{bmatrix}$$

$$B_J = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} = \begin{bmatrix} \hat{\underline{x}}_J & \hat{\underline{y}}_J \end{bmatrix}$$

$${}^{w}R_J = B_w^T B_J = I^T B_J = B_J = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$$

$\boxed{{}^{w}R_J = B_w^T B_J}$ extends to 3D as well

${}^{w}R_J$ is 3×3, B_w, B_J are also 3×3

Example HW or exam problem

Basis vectors

Set (1) $(\hat{\underline{u}}, \hat{\underline{v}}, \hat{\underline{w}})$ = $\hat{\underline{u}} = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$, $\hat{\underline{v}} = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$, $\hat{\underline{w}} = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$

Set (2) $(\hat{\underline{x}}, \hat{\underline{y}}, \hat{\underline{z}})$ $\hat{\underline{x}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\hat{\underline{y}} = \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$, $\hat{\underline{z}} = \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$

Find Rotation matrix that converts coordinates from (1) → (2)

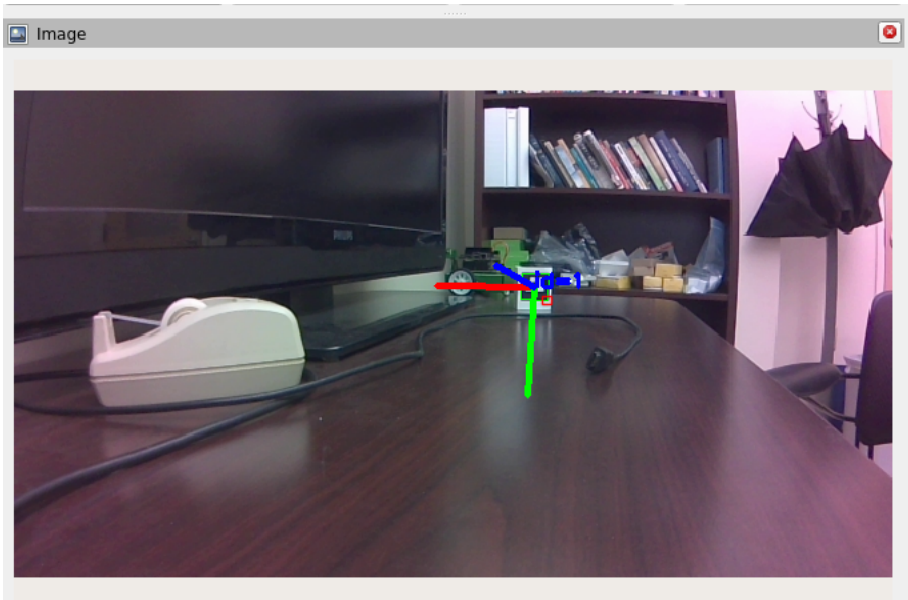
and then from ② → ①

$$ {}^{2}R_{1} = B_2^{T} B_1 $$

$$ = \begin{bmatrix} \hat{\underline{u}} & \hat{\underline{v}} & \hat{\underline{w}} \end{bmatrix}^{T} \begin{bmatrix} \hat{\underline{x}} & \hat{\underline{y}} & \hat{\underline{z}} \end{bmatrix} $$

$$ = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ -1/2 & -1/2 & -\frac{1}{\sqrt{2}} \\ +1/2 & +1/2 & -1/\sqrt{2} \end{bmatrix} $$

Axis-angle Representation/Quaternions



(3D) Rotation matrices (Basis matrices)

- (3D) Euler angles (Gimbal lock)
- Axis angle
- Quaternion

```
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aruco_detections
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Publisher count: 1
Subscription count: 0

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    nanosec: 924374790
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      z: 0.980432215194246
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      y: 0.9811997541144667
      z: -0.03227342049856023
      w: -0.18793966432455353
boards: []
---
```
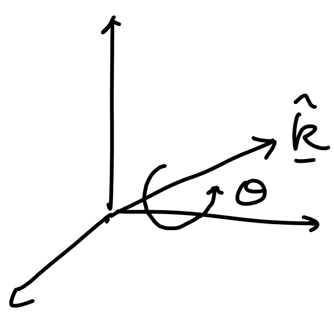

Quaternion

Axis-angle representation

$$\hat{\underline{k}} = \begin{bmatrix} k_x \\ k_y \\ k_z \end{bmatrix} \qquad \theta$$

unit vector



$$R \longleftrightarrow \hat{\underline{k}}, \theta$$

Quaternions

$$q = w + ix + jy + kz$$

$$\underbrace{i^2 = -1,\ j^2 = -1,\ k^2 = -1,\ ijk = -1}$$

$$q = \begin{bmatrix} w = \cos\left(\frac{\theta}{2}\right) \\ x = k_x \sin\left(\frac{\theta}{2}\right) \\ y = k_y \sin\left(\frac{\theta}{2}\right) \\ z = k_z \sin\left(\frac{\theta}{2}\right) \end{bmatrix}$$

$\underline{\hat{k}}, \theta$

axis, angle of rot

$$\sin\left(\frac{\theta}{2}\right) = \sqrt{x^2 + y^2 + z^2}, \quad \frac{\theta}{2} = \operatorname{arctan2}\left(w, \sqrt{x^2+y^2+z^2}\right)$$

$$k_x = \frac{x}{\sqrt{x^2+y^2+z^2}}, \quad \ldots\ldots$$

Axis-angle representation

$$(\hat{k}, \theta) \longrightarrow R$$

Rodrigues formula

$$\underline{y} = \underbrace{R(\hat{\underline{k}}, \theta)}\underline{x}$$

$$\boxed{\underline{y} = \underline{x} + \sin\theta\,(\hat{\underline{k}} \times \underline{x}) + (1-\cos\theta)\,(\hat{\underline{k}} \times (\hat{\underline{k}} \times \underline{x}))}$$

cross products

Cross product is a linear operation, hence it can be represented as a matrix multiplication

$$\underline{a} \times \underline{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = \hat{i}(a_y b_z - b_y a_z) - \hat{j}(a_x b_z - b_x a_z) \hat{k}(a_x b_y - b_x a_y)$$

$$\begin{bmatrix} a_y b_z - b_y a_z \\ -a_x b_z + b_x a_z \\ a_x b_y - b_x a_y \end{bmatrix} = \underline{a} \times \underline{b} = \underbrace{[\underline{a}_\times]}_{\text{cross product matrix}} \underline{b}$$

$$\underbrace{\begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix}} \underbrace{\begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix}}_{\underline{b}} = [\underline{a}_\times] \underline{b}$$

$[\underline{a}_\times]$ = cross product matrix is a skew-symmetric matrix $:= (A^T = -A)$

Rodrigues formula

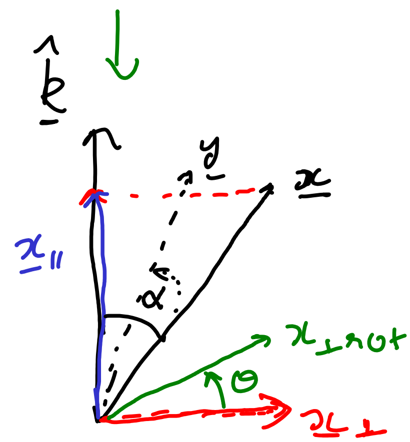
$$y = \underline{x} + \sin\theta (\underline{\hat{k}} \times \underline{x}) + (1-\cos\theta)(\underbrace{\underline{\hat{k}} \times (\underline{\hat{k}} \times \underline{x})}_{\neq (\hat{k} \times \hat{k}) \times \underline{x}})$$

$$y = \underline{x} + \sin\theta (\underbrace{[\underline{\hat{k}}_\times]}_{K} \underline{x}) + (1-\cos\theta)([\underline{\hat{k}}_\times][\underline{\hat{k}}_\times] \underline{x})$$

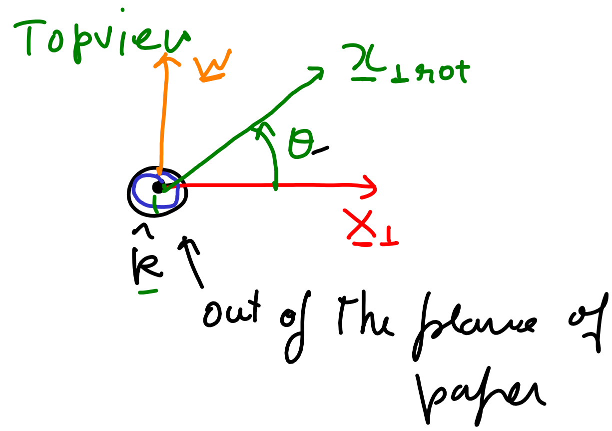
$$y = \boxed{(I_{3\times3} + \sin\theta K + (1-\cos\theta) K^2)} \underline{x} \qquad K^2 = KK$$

$$\boxed{R(\hat{k},\theta) = I_{3\times3} + K\sin\theta + (1-\cos\theta)K^2 \quad \text{where } K = [\hat{k}_\times]}$$

Derivation of the Rodrigues formula



$$\begin{array}{ccccc} x & = & x_{\parallel} & + & x_{\perp} \\ \downarrow R(\hat{k},\theta) & & & & \downarrow R(\hat{k},\theta) \\ y & = & x_{\parallel} & + & x_{\perp rot} \end{array}$$



$x_{\perp rot} = ?\ x_{\perp}$

$x_{\perp} = ?$

Observe that $\hat{k}$, x, $x_{\perp}$ are in the same plane.

w is $\perp$ to that plane

$w \propto \hat{k} \times x$

$x_{\perp}$ is $\perp$ to $\hat{k}$ and w

$x_{\perp} \propto w \times \hat{k}$

$x_{\perp} \propto -\hat{k} \times w = -(\hat{k} \times (\hat{k} \times x))$

$$|\underline{x}_\perp| = |\underline{x}|\sin\alpha$$

where α is the angle between $\hat{\underline{k}}$ and $\underline{x}$

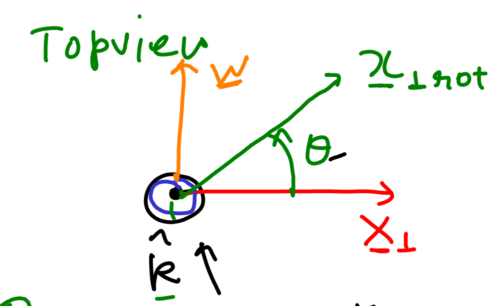
$$|\hat{\underline{k}}\times\underline{x}| = |\hat{\underline{k}}||\underline{x}|\sin\alpha$$
$$= |\underline{x}|\sin\alpha = |\underline{w}| = |\underline{x}_\perp|$$

$$|\hat{k}\times(\underbrace{\hat{k}\times x}_{\underline{w}})| = |\hat{k}||\hat{k}\times x|\sin 90^\circ$$
$$= |\hat{\underline{k}}\times\underline{x}| = |\underline{x}|\sin\alpha$$

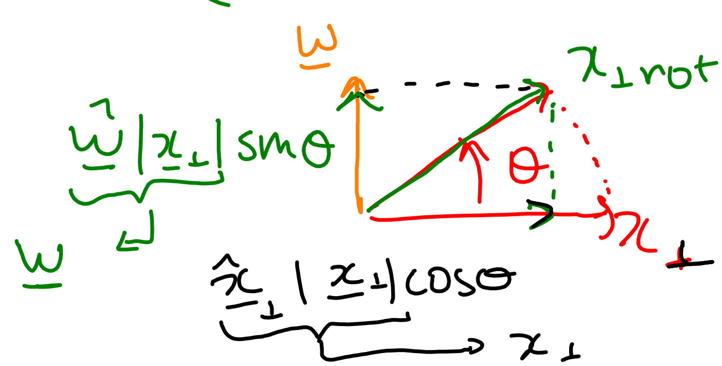
$$\underline{x}_\perp = -\hat{k}\times(\hat{k}\times\underline{x})$$

Rotate this by θ

$$|\underline{x}_\perp| = |x_{\perp rot}|$$



$$\underline{x}_{\perp rot} = \underline{w}\sin\theta + \underline{x}_\perp\cos\theta$$
$$= (\hat{\underline{k}}\times\underline{x})\sin\theta - \hat{k}\times(\hat{k}\times\underline{x})\cos\theta$$



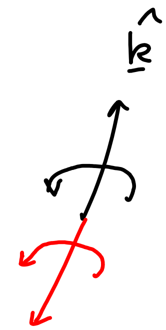
$$\underline{x}_{\parallel} = \underline{x} - \underline{x}_{\perp} \; .$$

$$\underline{y} = \underline{x}_{\parallel} + \underline{x}_{\perp rot}$$

$$= \underline{x} - \underline{x}_{\perp} + (\hat{\underline{k}} \times \underline{x}) \sin\theta - \underbrace{\hat{\underline{k}} \times (\hat{\underline{k}} \times \underline{x})}_{\perp} \cos\theta$$

$$= \underline{x} + (\hat{\underline{k}} \times (\hat{\underline{k}} \times \underline{x})) +$$

Rodrigues formula



3DOF, 4 numbers

$$\text{Axis-angle } (\hat{\underline{k}}, \theta) \longrightarrow \text{Rot matrix}$$

$$\text{Axis-angle } \xleftarrow{\text{2 solutions}} \text{Rot matrix}$$

$$R = I + \sin\theta \, K + (1 - \cos\theta) K^2$$

$$K_{3\times3} = [\hat{k}_\times] = \begin{bmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{bmatrix}$$

$$K^2 = \begin{bmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{bmatrix} \begin{bmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -(k_z^2 + k_y^2) & k_y k_x & k_z k_x \\ k_x k_y & -(k_x^2 + k_z^2) & k_z k_y \\ k_z k_x & k_z k_y & -(k_x^2 + k_y^2) \end{bmatrix}$$

$$R = I + \sin\theta\, K + (1-\cos\theta) K^2$$

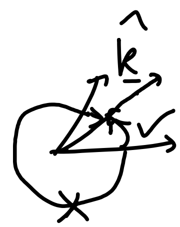
$$\overbrace{\begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{22} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}}^{R} = \begin{bmatrix} 1-(1-\cos\theta)(k_z^2+k_y^2) & (1-\cos\theta) k_y k_x - \sin\theta\, k_z & \\ & 1-(1-\cos\theta)(k_x^2+k_z^2) & \\ & & 1-(1-\cos\theta)(k_z^2+k_y^2) \end{bmatrix}$$

$$\mathrm{trace}(R) = r_{11} + r_{22} + r_{33} = 3 - (1-\cos\theta)\underbrace{(k_x^2+k_y^2+k_z^2)}_{1}\, 2$$

$$= 3 - 2 + 2\cos\theta$$

$$= 1 + 2\cos\theta$$

$$\Rightarrow \boxed{\theta = \cos^{-1}\left(\frac{\mathrm{trace}(R) - 1}{2}\right)} \quad \text{angle} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$



if $\theta = 0$ or $\theta = 180^\circ$

$$\hat{k} = \begin{pmatrix} k_x \\ k_y \\ k_z \end{pmatrix} = \begin{bmatrix} \pm\sqrt{(r_{11}+1)/2} \\ \pm\sqrt{(r_{22}+1)/2} \\ \pm\sqrt{(r_{33}+1)/2} \end{bmatrix}$$

$$\sin\theta = \overbrace{0}^{180^\circ}$$

$$\cos\theta = -1$$

$$R = I + 2K^2$$

$$R = \begin{bmatrix} 1-k_z^2-k_y^2 & & \\ & 1-k_x^2-k_z^2 & \\ & & 1-k_x^2-k_y^2 \end{bmatrix}$$

$$k_x^2 + k_y^2 + k_z^2 = 1 \Rightarrow k_x^2 = \underbrace{1-k_y^2-k_z^2}$$

$$R = \begin{bmatrix} k_x^2 & 1+2k_y k_x & \\ & k_y & \\ & & k_z^2 \end{bmatrix}$$

else $\theta \neq 0, 180^\circ$

$$\hat{k} = \begin{bmatrix} k_x \\ k_y \\ k_z \end{bmatrix} = \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} \frac{1}{2\sin\theta}$$

Geometric interpretation of Eigen vector and Eigen values

$$R\underline{v} = \lambda \underline{v} \leftarrow \text{solutions} \rightarrow \underline{v} \neq \underline{0},\ \lambda \neq 0$$

Eigen vectors and values of R

- Real eigenvalue, $\lambda = 1$, $\underline{\hat{k}} \leftarrow$ eigen vector
  - $\underline{v}$, $\lambda \underline{v}$, $\lambda \underline{v}$
- Complex
- Complex

Rotation matrix preserves vector lengths

Orthonormal

$R^{-1} = R^T$

① $R^T R = I$

② $\det(R) = +1$

$$\underline{y} = R\underline{x} \xrightarrow{\text{Proof}} \|\underline{y}\| = \|x\|$$

Use: $\|\underline{y}\|^2 = \underline{y}^T \underline{y}$

$$\|y\|^2 = \underline{y}^T \underline{y} = (R\underline{x})^T (R\underline{x}) = \underline{x}^T \underbrace{R^T R}\, \underline{x} = \underline{x}^T I \underline{x} = \underline{x}^T \underline{x}$$

$$= \|x\|^2$$